

ALI AJVAND

Data Scientist

Forecasting & Risk Modeling · Machine Learning · Business Intelligence

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PROFESSIONAL SUMMARY

Data Scientist who builds forecasting, risk, and BI systems that replace an existing manual process and can prove it. I validate chronologically, audit for leakage before claiming a result, and benchmark against the incumbent baseline rather than another model: **9.36% WMAPE** demand forecasts against a 34.06% planner heuristic, and a calibrated credit scorecard that cut defaults on the approved book by **38%** at a higher approval rate. 2nd of 13 shortlisted teams at Hackathon Pol 4 (AI Innovation Factory × Jobama). Comfortable owning the full path from SQL to deployed API to the dashboard a manager actually opens.

PROFESSIONAL EXPERIENCE

Data Analyst (Project-Based) — BehinBazar | client: JTI (Japan Tobacco International) Tehran, Iran · Apr 2025 – Nov 2025

- Built the ETL pipelines and Power BI layer behind JTI's Iran field-sales operations, replacing a multi-day reporting lag with a same-day view of sales performance, territory coverage, and team productivity.
- Automated weekly reporting through Python and the Google Sheets API: **40% less manual effort** and no more copy-paste errors in numbers management acted on.
- Built Folium geospatial tooling for store-coverage analysis and route planning, turning ad-hoc visit scheduling into a repeatable process field managers ran themselves.
- Consolidated scattered team progress updates into one report, which fixed data-quality gaps feeding every downstream dashboard.

SELECTED DATA SCIENCE & MACHINE LEARNING PROJECTS

Demand & Sales Forecasting — Hierarchical Retail Time-Series Pipeline 2026

[View code & results ↗](#)

- Cut forecast error **72%** versus the planner's same-weekday heuristic (34.06% → 9.36% WMAPE), which moved **98.3% of 1,115 stores** inside the 15% threshold where replenishment can run without human review.
- Forecasted 844,338 open store-days across three non-overlapping 42-day walk-forward folds, with training data always ending before the fold begins, and held bias to -1.9% against a $\pm 5\%$ target.
- Published per-store error (median 8.7%, p90 11.9%) so safety stock could be set store by store instead of one padded number chain-wide.

Credit Risk Decision Engine — Calibrated PD Scorecard & Approval Policy 2026

[View code & results ↗](#)

- Cut the bad rate on the approved book from 15.9% to **9.8%** (a 38% relative reduction in defaults) while approving 54.4% of applicants, by optimizing a 5:1 false-approval-to-false-decline cost policy instead of a naive $PD < 0.50$ cutoff.
- Calibrated PD across 30,000 clients on 41 behavioral features: ROC-AUC 0.780, PR-AUC 0.554, **KS 0.427**, Brier 0.135, against 0.703 for the single-rule baseline; the riskiest decile defaults at 70.0% versus 22.1% portfolio-wide (3.17× lift).
- Kept sex, education, marital status, and age out of the feature matrix entirely, retaining them only for adverse-impact monitoring, and shipped a threshold table the business can re-cost when loss assumptions change.

EV & Automotive Analytics — Market Intelligence & Price Regression 2026

[View code & results ↗](#)

- Caught a leakage failure before it shipped: an apparent ROC-AUC of 1.00 collapsed to 0.66 once drivetrain-derived proxy features were removed, killing a model claim that would not have survived production.
- Priced 15,409 used-car listings on a nameplate-grouped split, where an explainable linear regression (**R^2 0.83, 19.3% MAPE**) beat gradient boosting at 20.8% and more than halved the 41.8% error of the brand-median rule pricing teams use by hand.
- Modeled 289,555 EV registrations in a Power BI star schema: 80.7% BEV share and 69.3% fleet concentration in the top three counties, pointing charging-infrastructure spend at where the fleet actually is.

Olist Delivery Performance Analytics — Predictive Logistics & Decision Policies 2026

[View code & results ↗](#)

- Predicted late delivery, freight cost, and promise dates across ~99,000 Brazilian e-commerce orders (3,000 sellers, 27 states) using purchase-time information only, so every model is deployable at checkout rather than after the fact.
- Engineered seller, route, and customer-history features through as-of joins keyed to purchase time, then proved the pipeline clean with a 39-feature leakage audit and 34 automated tests.
- Turned five models into policies operations can run: capacity-constrained expedite queues and quantile promise dates, each with isotonic calibration, cost-sensitivity tables, and an incumbent baseline to beat.

Hackathon Pol 4 — Hospitality Demand Forecasting (2nd Place)

AI Innovation Factory × Jobama · 72-hour competition, 13 shortlisted teams · 2026

- Placed **2nd of 13 shortlisted teams** with a time-series ensemble (LightGBM, XGBoost, CatBoost, HistGradientBoosting) hitting **82% accuracy** on real hotel booking data across 321 Iranian cities, built from scratch in 72 hours.
- Combined accrual-based nowcasting with calendar seasonality and geographic features, then framed the output as an early-warning signal: when to move inventory and pricing 2–4 weeks ahead of a demand peak.

TECHNICAL SKILLS

Languages & Data: Python, SQL, DAX, pandas, PostgreSQL, MySQL, SQL Server, DuckDB, Parquet, ETL/ELT

Machine Learning: scikit-learn, XGBoost, LightGBM, CatBoost, forecasting, classification, regression, clustering, feature engineering, SHAP, probability calibration (isotonic, Platt)

Deep Learning & NLP: PyTorch, Hugging Face Transformers, Sentence-Transformers, FAISS, LangChain (LLM orchestration & retrieval pipelines)

Statistics & Validation: walk-forward and grouped cross-validation, leakage auditing, A/B testing, hypothesis testing, cost-sensitive evaluation

BI & Visualization: Power BI (star schema, DAX, drill-through, KPI dictionaries), Plotly, Seaborn, Matplotlib, Folium

Engineering & MLOps: FastAPI, Docker, Git, GitHub Actions CI/CD, pytest, ruff, mypy, reproducible pipelines

Also familiar: Tableau, Metabase, Flask, Streamlit, T-SQL

ADDITIONAL WORK

LangChain + FAISS retrieval-augmented generation pipeline over structured vehicle data (Hugging Face embeddings) · RAG clinical decision-support API (FastAPI, FAISS, Sentence-Transformers, PyTorch) · Power BI e-commerce sales dashboard on MySQL · K-means customer segmentation · FP-Growth market-basket analysis · A/B testing & ad-spend analysis. [Full case studies ↗](#)

EDUCATION & CERTIFICATIONS

B.Sc. Computer Engineering — Islamic Azad University, Science & Research Branch

Tehran · 2026

Graduated 2026 · GPA 16/20 · Coursework: data structures & algorithms, databases, statistics & probability, linear algebra, software engineering.

M.Sc. entrance examination completed (2026) — program placement pending.

Certifications: Data Scientist Bootcamp, Daneshkar (100+ hrs, 2025) · LangChain for LLM Application Development, freeCodeCamp (2026) · Business Intelligence Bootcamp, Udemy (2023) · Data Analyst Bootcamp, Alex The Analyst (2023)

HONORS & LANGUAGES

2nd Place of 13 shortlisted teams — Hackathon Pol 4, AI Innovation Factory × Jobama, demand forecasting, 2026. · **Languages:** Persian (native), English (professional working proficiency).